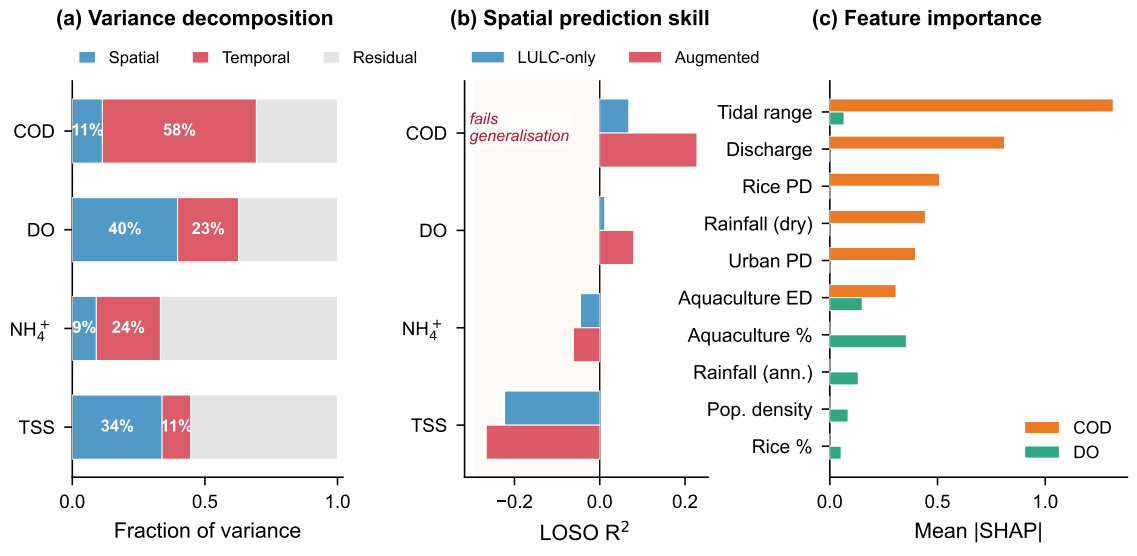
(a) Variance decomposition
Spatial
Temporal
Residual
COD
DO
NH4+
TSS
11%
58%
40%
23%
9%
24%
34%
11%
0.0
0.5
1.0
Fraction of variance
(b) Spatial prediction skill
LULC-only
Augmented
fails generalisation
COD
DO
NH4+
TSS
−0.2
0.0
0.2
LOSO R2
(c) Feature importance
Tidal range
Discharge
Rice PD
Rainfall (dry)
Urban PD
Aquaculture ED
Aquaculture %
Rainfall (ann.)
Pop. density
Rice %
COD
DO
0.0
0.5
1.0
Mean |SHAP|